# Jacob Foster

+1 289-208-8765 fostej26@mcmaster.ca in/jacobnfoster fostej26.github.io github.com/fostej26

## Education

**McMaster University, Hamilton, ON** **Sep. 2022 - May 2027**
*Bachelor of Mechatronics Engineering* *cGPA - 3.86/4.00*

- **Relevant Coursework:** Data Structures and Algorithms, Embedded System Design, Analog and Digital Circuits, Software Development and Design, Operating Systems, FPGA Design.
- Accolades: Faculty of Engineering Award of Excellence Scholarship 2022, Dean's Honours List 2022-2024.
- Involvement: McMaster Solar Car Project Algorithms Developer, McMaster SumoBot 2024 Group Champion, Deltahacks X, Deltahacks XI, McMaster CAD Designathon, Google Developer Student Club Member.

## Experience

**L3Harris Technologies - Wescam Division** **May 2024 - August 2024**
*Manufacturing Engineering Co-op* *Waterdown, ON*

- Developed a software/firmware comparison web tool to capture current and previous build components of a WESCAM imaging device and automatically determine build status for repairs, **decreasing effective labour time from 20 mins to 1 min**. Built using **C#, .NET Framework, HTML, CSS.**
- Automated several test processes using **C#** to load, save, and unload calibration values onto devices and into test directories.
- Implemented an improved tooling management system for a lab of **100+ manufacturing technicians** using housing assemblies designed in **SOLIDWORKS.**

**McMaster Solar Car Project** **October 2024 - Present**
*Race Strategies Algorithm Developer* *Hamilton, ON*

- Develop responsive algorithms using **Python and Scikit Learn ML** to compute optimal travel speed and conserve energy based on solar irradiance, geological, and social factors.
- Modularize and document previous race strategies codebase using **SDLC and OOP principles.**

## Projects

**Pocket Spotter - Virtual Fitness Assistant** | *Python, OpenCV, Flask, Javascript, HTML, CSS* **January 2025**

- Developed a **Python and Flask web app** to determine imperfections in a user's form for various exercises in real time.
- Created mathematical functions to detect significant muscular fatigue, strength imbalances, and incorrect motion paths that may cause injury when performing exercises based on timings and joint angles captured with **OpenCV**.
- Implemented a system where all performed sets of a given exercise are tracked using and presented in a summary at the end of a workout.

**AllergyPal** | *Python, Tensorflow, HTML, CSS, Javascript, Flask* **September 2024**

- Designed a website that allows users input a list of dietary restrictions and use machine learning to detect potential hazards in their food.
- Trained a **machine learning model to detect over 100 different foods** using images taken by the user in the site.
- Utilized the ChatGPT 4.0 API to retrieve common ingredients and food risks present in any given dish, which was then cross referenced with a user's dietary restrictions.
- Created a **REST API using Flask and Python** to call upon the image recognition model in real time upon a user's request.

**Spinnick SumoBot** | *Arduino, C++, Autodesk Inventor, Embedded Systems, Soldering* **January 2024**

- Engineered and automated battle robot capable of perceiving environment features, attacking opponents, and evading attacks.
- **Developed C++ control algorithms for attack and evasion** based on various sensor readings.
- Assembled the bot's control mechanism using an **Arduino micro-controller, ultrasonic and colour sensors, and DC motors.**
- CAD designed and printed the bot's body using Autodesk Inventor.

## Technical Skills

**Computing**: Python, C, C++, C#, HTML, CSS, Javascript, Tailwind, React.js, MATLAB, R
**Developer Tools**: Git, GitHub, Linux, .NET, Tensorflow, VS Code, Visual Studio, Node.js, GDB, Google Cloud
**Technologies**: SOLIDWORKS, Autodesk Inventor, LaTeX, Adobe Photoshop, Microsoft Office
**Manufacturing**: Soldering, 3D printing, Drill press, Hand and power tools